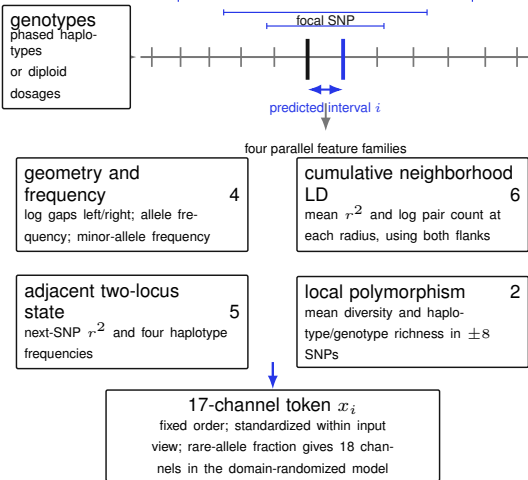
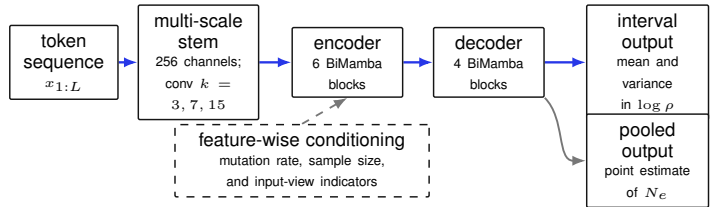


a Mapping variant to interval tokens LD radii 5, 25, 50 kb

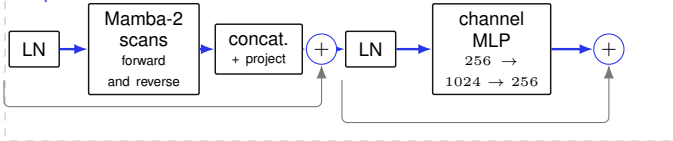


Phased input uses haplotype LD. Unphased input substitutes composite dosage LD; minor-allele folding removes dependence on ancestral-state polarization.

b Bidirectional state-space inference



one pre-normalized BiMamba block



Two selective passes collect left and right LD information in the linear time sequence. Longer sequences ($> 1,024$ tokens) are scanned in 25% overlap with first and second moments of stitching. The trained checkpoint is then frozen.